

Homework 1 - Foundations of proof

Five questions. Each question is worth 10 points (50 total).

Write a clear argument when requested. Valid alternative proofs and clear detours are welcome. Follow each exercise's specific requirements. This handout contains no solutions.

Question 1. Sets in three forms

Let $\mathbb{N}=\{0,1,2,\dots\}$. Give short answers; no justification is required.

(a) List $A=\{n\in\mathbb{N}:12<n<90 \text{ and } n \text{ is a square or a cube}\}$ in roster notation.

(b) Write $B=\{1,6,11,16,\dots\}$ in set-builder notation.

(c) Write $C=\{\dots, 1/27, 1/9, 1/3, 1, 3, 9, 27, \dots\}$ in set-builder notation.

Question 2. A divisibility claim

Determine whether the proposition is true or false. Prove it if true; otherwise give and verify a counterexample.

For all integers a,b , if $10 \mid ab$, then $10 \mid a$ or $10 \mid b$.

Question 3. A rational point inside an interval

Let $x,y\in\mathbb{Q}$ with $x<y$, and let $u=(3x+y)/4$. Prove that $u\in\mathbb{Q}$ and $x<u<y$.

For rationality, use the definition of a rational number; merely citing closure of $\mathbb{Q}$ is not sufficient for this exercise.

Question 4. An odd expression

Prove that n^2+3n+1 is odd for every integer n . Your argument must cover negative integers as well as nonnegative integers.

Question 5. Removing either set

For arbitrary sets A, B, C , prove by double containment that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.